BHUSATYA

Product Requirements Document

Fake Land Document & Tampering Detector

Field	Details
Product	BhuSatya
Category	GovTech / AI-assisted document verification
Platform	Web application
MVP	Upload → extraction → forensic analysis → verification report
Core principle	Assist human verification with evidence; do not blindly declare documents fake.

1. Product Vision

BhuSatya helps citizens, banks, legal professionals and authorized land-record verifiers examine land documents for suspicious alterations, inconsistencies and possible record mismatches.

The system analyzes documents at three levels:

```
DOCUMENT
   ↓
CONTENT
   ↓
FORENSICS
   ↓
OFFICIAL RECORDS
   ↓
EVIDENCE-BASED REPORT
```

BhuSatya makes initial verification faster and easier to investigate while keeping final legal or official verification with an authorized human.

2. Problem Statement

Land documents can contain subtle manipulations that are difficult to identify during manual inspection.

- Altered names or land details
- Modified survey numbers or land area
- Fake or copied stamps and seals
- Modified or substituted signatures
- Spliced or manipulated images
- Modified PDF text layers
- Altered survey or map boundaries

Verification is harder when records are handwritten, old, low-quality, scanned, image-based, or written in regional languages. These challenges affect farmers, property buyers, banks, legal professionals and government authorities.

3. Target Users

User	Primary need
Citizen / Farmer	Understand whether a document contains suspicious inconsistencies.
Bank / Financial Institution	Screen property documents before loans or collateral decisions.
Legal Professional	Identify suspicious fields and potentially manipulated sections quickly.
Authorized Government / Registry Verifier	Assist with document screening, comparison and investigation.

4. Core User Journey

1. Login
2. Open dashboard
3. Upload land document
4. Preview and select/confirm document type

5. Start verification
6. OCR and data extraction
7. Forensic analysis
8. Field consistency check
9. Cross-check supported official records
10. Generate verification report
11. Inspect evidence
12. Download/share report

5. Application Structure

```
BhuSatya
├── Dashboard
├── Documents
│   ├── All Documents
│   ├── Pending
│   ├── Verified
│   └── Requires Review
├── New Verification
├── Reports
└── Settings
```

6. Dashboard Requirements

- Overview cards: Total Documents, Verified, Requires Review, Anomalies Detected
- Recent documents list
- Document type, upload date and status
- Verification confidence indicator
- Simple verification activity visualization
- Quick action to start a new verification

7. New Verification Flow

7.1 Upload

- Drag-and-drop or browse upload
- Supported formats: PDF, JPG, JPEG, PNG
- File-size guidance
- Continue after successful upload

7.2 Document Type

- Registry / Sale Deed
- Khasra-Khatauni
- B-1
- Naksha
- Power of Attorney
- Other
- Optional auto-detection

7.3 Preview & Analysis Options

- Document preview
- File name, page count, language, document type, quality and upload date
- OCR & Data Extraction
- Tampering Detection
- Field Consistency Check
- Registry Cross-Check

8. Processing Experience

Show meaningful processing stages instead of a generic loading indicator.

```
✓ Document received
✓ Document classification
✓ OCR extraction
● Forensic analysis
○ Field consistency
○ Registry verification
○ Report generation
```

9. Verification Report — Core Product Screen

The Verification Report is the most important screen. It should make evidence understandable within seconds.

Overall Assessment

- Use statuses such as Verified, Requires Verification, or High-Risk / Anomaly Detected.
- Show verification confidence.
- Show the number of findings requiring closer inspection.

Verification Breakdown

Area	Example result
Document Integrity	92% — No major structural anomalies detected
Owner Name	Match
Survey Number	Match
Land Area	Mismatch
Stamp	Possible anomaly
Signature	No strong anomaly
Text Layer	Inconsistency detected
Registry Comparison	Requires review

BhuSatya should communicate evidence and recommend human review rather than present an AI prediction as legal proof that a document is fake.

10. Evidence Viewer

- Original document view
- Highlighted suspicious region
- Finding name
- Reason for finding
- Confidence indicator
- Recommendation such as “Manual verification recommended”

11. Extracted Information

- Owner name
- Father's name where present
- Survey number
- Village
- District
- Land area
- Registration date
- Other document-specific fields

Each field can show Match, Mismatch or Low OCR Confidence.

12. Registry Comparison

Where supported official records or APIs are available, extracted fields can be compared against the corresponding record.

```
DOCUMENT                OFFICIAL RECORD
Owner: Ramesh Kumar  =  Ramesh Kumar      ✓
Survey: 124/2        =  124/2             ✓
Land Area: 2.50 ha   ≠  2.18 ha           ⚠
Village: Rampur      =  Rampur            ✓
```

For the hackathon MVP, a controlled or mock registry dataset can demonstrate this flow if a live government API is unavailable.

13. Reports

- Filter by All, Verified, Review and Anomaly
- Show document name, date, status and confidence
- Open report
- Download PDF

14. Generated PDF Report

- BhuSatya header and document ID
- Document type, pages and analysis date
- Overall assessment
- Verification findings
- Evidence thumbnails / highlighted regions
- Registry comparison where available
- AI-assisted screening disclaimer

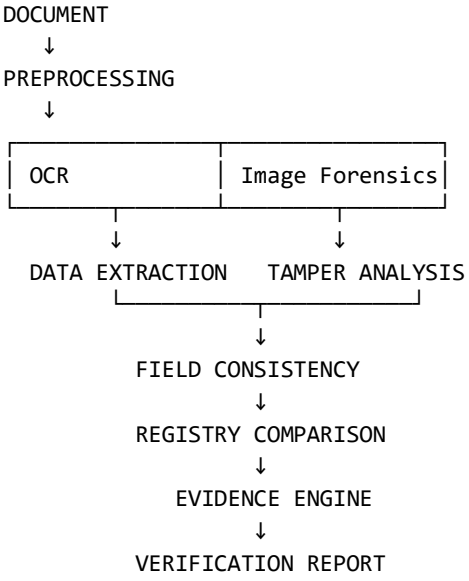
Suggested disclaimer: “This report is an AI-assisted screening result and does not constitute legal proof of fraud or document authenticity. Final verification should be performed by an authorized authority.”

15. Settings

- Profile and organization details
- Role

- Language: English / Hindi
- Password and authentication settings
- Two-factor authentication where supported
- Notification preferences

16. AI / Processing Architecture



17. Suggested Technology Stack

Layer	Suggested technology
Frontend	Next.js + React + Tailwind CSS
Backend	FastAPI
OCR	PaddleOCR / Tesseract or suitable document OCR
Image / Document Analysis	Python + OpenCV + PDF parsing
AI / Document Understanding	Suitable vision/document model
Database	PostgreSQL or lightweight prototype database
File Storage	Secure object/file storage

18. Roles & Permissions

Role	Capabilities
Citizen	Upload documents, view own reports, download reports
Organization / Bank	Upload documents, manage cases, view reports
Verifier	Inspect documents, review anomalies, compare records, flag cases

19. Security Requirements

- Authentication and role-based access
- Encrypted file transfer
- Secure document storage
- Access control
- Audit logs

- Session timeout
- Minimize exposure of uploaded documents
- Uploaded documents should not automatically become training data for the AI system

20. MVP Scope

The hackathon MVP should focus on one complete, demonstrable verification workflow.

13. Upload a land document
14. Run OCR
15. Extract key fields
16. Detect a limited set of meaningful tampering patterns
17. Compare extracted fields against a controlled/mock registry dataset
18. Generate an explainable verification report

Do not try to solve nationwide portal integration, every regional language, perfect handwriting recognition or legal authenticity certification in the first version.

21. Recommended Hackathon Demo Scenario

Use one realistic sample document throughout the demo so judges can follow the complete journey.

```
Registry_1242.pdf

Extracted:
Owner: Ramesh Kumar
Survey No.: 124/2
Land Area: 2.50 hectares
Village: Rampur

Registry record:
Owner: Ramesh Kumar        ✓
Survey No.: 124/2          ✓
Land Area: 2.18 hectares   ⚠
Village: Rampur             ✓

Forensic findings:
Stamp region: suspicious
Text layer: inconsistent

Final assessment:
REQUIRES VERIFICATION
```

The judge should be able to open findings and see the corresponding evidence on the document.

22. Design Direction

- Serious GovTech verification-platform feel
- Neutral warm/off-white or light background
- Deep earthy brown or dark navy primary
- Terracotta accent
- Muted green for positive results
- Amber for warnings

- Restrained red for serious anomalies
- Inter, DM Sans or IBM Plex Sans
- Document imagery, forensic highlights, map outlines and annotation lines
- Avoid generic AI robots, glowing gradients and excessive decorative cards

23. Product Success Criteria

- A first-time user can upload and start verification without assistance.
- The system clearly communicates what it is checking.
- Extracted information is easier to understand than raw OCR output.
- Suspicious findings are linked to visible evidence.
- Registry mismatches are easy to compare.
- The report distinguishes AI-assisted screening from legal/official determination.
- The complete MVP flow can be demonstrated end-to-end during the hackathon.

BHUSATYA

Verify the document before you trust it.